

Migration Guide

A CEREBRUM Research Publication (v1.2.4)

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Abstract

Overview: This guide provides a conceptual entry point into the CEREBRUM framework. It is designed for practitioners and researchers seeking to understand the core reasoning signals of Framework Implementation Guide.

0.1 CEREBRUM Migration Guide

This guide covers what operators and developers need to **do** (not just what changed) when upgrading between major versions. For a list of *what* changed, see `CHANGELOG.md`.

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v1.0.0 → v1.1.0 (Phase 20: Relativistic Hardening)

Required actions: none

All four Phase 20 fixes use backward-compatible defaults. No existing code breaks.

Recommended changes

1. Enable Query Snapshot Isolation (Hole 5)

If your deployment uses the `GlobalRebalancer` for live graph updates alongside concurrent queries, enable snapshot isolation explicitly:

```
# Before (v1.0.0      no snapshot)
traversal = BeamTraversal(adapter=adapter, csa_engine=csa, ...)
```

```
# After (v1.1.0      add snapshot at query start)
csa.set_query_snapshot(adapter.community_map)
traversal = BeamTraversal(adapter=adapter, csa_engine=csa, ...)
```

The `api/server.py` already does this automatically for REST queries.

2. Per-community CSA parameters for homogeneous graphs (Hole 6)

If your graph has tightly-clustered domains (proteins, legal clauses, financial instruments) where intra-community similarity is uniformly high, add parameter overrides for those communities:

```
# Identify homogeneous communities (avg intra-community S_C > 0.85)
# Then reduce and increase / for those communities:
csa = CSAEngine(
    community_params={3: (0.5, 0.15, 0.25, 0.05, 0.05, 0.0)},
    ...
)
```

3. Path-preserving hold-out is now the default

`InferenceValidator` now defaults to `path_preserving=True`. On dense graphs this is a no-op. On sparse graphs (avg degree <3) this will improve recall estimates. No code changes needed unless you specifically want the old behavior:

```
# To restore v1.0.0 behavior (not recommended):
validator = InferenceValidator(adapter, traversal, path_preserving=False)
```

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v0.4.0 → v1.0.0 (Phase 19: Production Hardening)

Required actions

1. Namespace isolation for mixed-modality deployments

If you use both `IngestionPipeline` and `SignalEncoder` in the same graph, you must now add namespace prefixes to prevent entity collisions:

```
# Before (v0.4.0      collision risk)
pipeline = IngestionPipeline(adapter)
encoder  = SignalEncoder(entity_dim=128)

# After (v1.0.0      isolated namespaces)
pipeline = IngestionPipeline(adapter, namespace="text")
encoder  = SignalEncoder(entity_dim=128, namespace="signal")
```

If your graph contains entities with the same name in both text and signal sources, they will now appear as separate nodes (`text:Name` and `signal:Name`). To intentionally merge them, use `entity_dedup_map="signal:text:Name"` in `IngestionPipeline`.

2. Wire the Zombie Bridge hook if using `GlobalRebalancer` + `BridgeTwinEngine`

```
# Before (v0.4.0      stale bridges after rebalance)
rebalancer = GlobalRebalancer(adapter, q_drift_threshold=0.05)

# After (v1.0.0      bridges stay consistent)
rebalancer = GlobalRebalancer(adapter, q_drift_threshold=0.05, bridge_engine=
    bridge_engine)
```

Recommended changes

3. Enable warm-start for probabilistic traversal on sparse graphs

```
# Before
traversal = BeamTraversal(adapter, csa, probabilistic=True)

# After      reduce cold-start variance 85%
traversal = BeamTraversal(adapter, csa, probabilistic=True,
    warm_start_strength=5.0)
```

4. Enable causal flood protection for untrusted event streams

```
# Before
discretizer = STDPDiscretizer(w_threshold=0.5, n_min=5)

# After      adversarial hardening
```

```
discretizer = STDPDiscretizer(w_threshold=0.5, n_min=5, min_causal_span=1.0,
    use_chi_squared=True)
```

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v0.3.0 → v0.4.0 (Phase 18: v0.4 Horizon)

New required environment variable (if using API auth)

```
export CEREBRUM_JWT_SECRET="your-secret-key"
```

The API server now requires JWT authentication on all endpoints. Anonymous access can be re-enabled for development:

```
export CEREBRUM_ALLOW_ANONYMOUS=true
```

IngestionPipeline replaces direct adapter.add_edge for text ingest

```
# Before (v0.3.0)
for subj, pred, obj in triples:
    adapter.add_edge(subj, obj, pred)

# After (v0.4.0)      preferred for text data
pipeline = IngestionPipeline(adapter)
for subj, pred, obj in triples:
    pipeline.process(subj, pred, obj, confidence=0.9)
```

Direct add_edge still works — the pipeline is additive.

LLM bridge import path changed

```
# Before
from cerebrum.bridge import generate

# After
from llm_bridge import generate, GenerationResult
```

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v0.2.0 → v0.3.0 (Phase 10–11: Production Hardening + Streaming)

igraph / leidenalg dependency removed

The native Leiden reimplementation (`core/leiden_native.py`) is now the default backend. Remove these from your environment if installed:

```
pip uninstall leidenalg python-igraph igraph
```

The native implementation is API-compatible; no code changes are needed.

JWT authentication added to API

See v0.3.0 → v0.4.0 section above (authentication was introduced in v0.3.0).

StreamAdapter replaces manual polling loops

```
# Before (v0.2.0      polling)
while True:
    event = get_next_event()
    adapter.add_edge(event.subject, event.object, event.predicate)
    time.sleep(0.1)

# After (v0.3.0      StreamAdapter)
stream = StreamAdapter(base_adapter=adapter, window_size=10_000)
stream.push_event(subject, predicate, object, timestamp, weight)
```

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v0.1.0 → v0.2.0 (Phase 6–9: Federated)

No breaking changes. All Phase 6–9 features (FederatedAdapter, HolographicIndex, Handshake) are additive and do not affect single-graph deployments.

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